NUS-H_2: Nuclear-Upgraded Steam Methane Reforming with Carbon Capture for Low-Carbon Hydrogen in Singapore

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Executive Summary

Singapore imports all of its natural gas, and conventional steam methane reforming (SMR) emits CO_2 from both its furnace and its reforming chemistry. NUS-H_2 removes the furnace. A 210 MW_{th} high-temperature gas-cooled reactor (HTGR) heats the reformer through helium, and 30 wt% monoethanolamine (MEA) captures CO_2 from shifted syngas. The plant is a new build at SOXAL scale on the proposed Western Island, a siting proposal of this study with no official approval. It makes 100,000 $Nm^3\ h^{-1}$ of purified hydrogen, 66,925 t $H_2\ y^{-1}$ at 85% availability, from 120.8 MW_{th} of process heat and a reactor need of 178.5 MW_{th}. Pressure swing adsorption (PSA) stays an external block, and its reject returns to the reformer as a stream crossing the boundary.

Net abatement is 0.578 $MtCO_2e\ y^{-1}$ against a same-size unabated SMR, which meets the 0.25 $MtCO_2e\ y^{-1}$ target by a factor of 2.3. The S$100/$tCO_2e$ cost target is missed in all 8 central cases. The base full cost is S$189.7/$tCO_2e$ at 25 y and S$165.6/$tCO_2e$ at 60 y for a build between first and Nth of a kind (BOAK), and S$277.9/$tCO_2e$ and S$241.7/$tCO_2e$ for a first of a kind (FOAK). A discounted cash flow (DCF) reproduces these within 0.2%. The levelised cost of hydrogen is S$5.26/kg at BOAK, 25 y, against S$3.62/kg for the unabated SMR.

A named scenario, MATURE_60Y, states what it would take to meet the target. A tenth-of-a-kind reactor on INL's BOAK learning route over 60 y, with gas at S$20.7/GJ against the central S$15/GJ, gives S$95.6/tCO_2e on the ledger and S$95.9/$tCO_2e$ by DCF. That gas price lies above the S$10 to S$20/GJ range tested, and at S$20/GJ the scenario gives S$99.6/$tCO_2e$. Dearer gas lowers the abatement cost because NUS-H_2 burns less gas than the unabated plant. Deployment towards 2050 depends on a qualified reactor and intermediate heat exchanger (IHX), a design-specific emergency planning zone (EPZ), a licensing framework and a contracted CO_2 storage pathway.

1. Problem and Proposed Solution

1.1 The problem and its Singapore context

Singapore's energy transition requires reliable industrial energy while reducing carbon emissions. With no domestic natural gas resources, Singapore relies on imported pipeline gas and liquefied natural gas (LNG); natural gas accounted for approximately 94% of electricity-generation fuel in 2024 [1,2]. This dependence highlights the strategic importance of diversifying low-carbon energy supply, particularly for continuous high-temperature industrial processes.

Hydrogen offers a pathway for decarbonising industrial applications and supporting Singapore's future energy system. The National Hydrogen Strategy (October 2022) identifies low-carbon hydrogen as a potential contributor to up to half of the country's power needs by 2050 [3]. SMR is an established hydrogen-production technology; it generates CO_2 through methane conversion and

conventionally burns fossil fuel to supply the endothermic reforming duty [4]. Addressing both emissions sources creates an opportunity to integrate nuclear process heat with carbon capture and storage (CCS).

CCS captures the CO_2 generated in the reforming train [5], while nuclear process heat displaces natural gas combustion for the reforming duty [6]. Steam reforming of natural gas operates within approximately 500 to 900°C (Figure 1) [7], a range that HTGRs can serve and light-water reactors cannot.

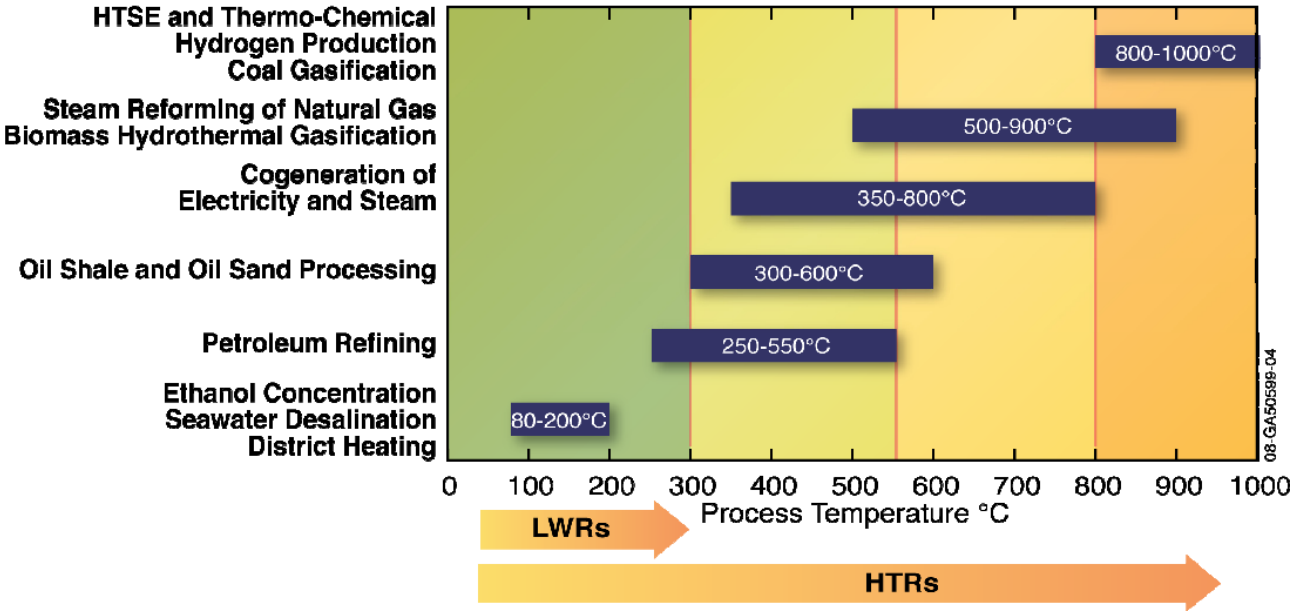


Figure 1. Process heat temperature ranges of industrial applications, with the ranges that light-water reactors (LWRs) and HTGRs can serve [7].

This report proposes NUS-H_2, a single-train HTGR-heated SMR integrated with MEA carbon capture, as a new build on the proposed Western Island. The site is a proposal of this study and has no official approval. The plant is sized on the SOXAL hydrogen plant on Jurong Island, about 100,000 $Nm^3\ h^{-1}$ [8,9], and is compared with a same-size unabated SMR delivering equal hydrogen in equal hours. The design targets are net abatement above 0.25 $MtCO_2e\ y^{-1}$ within Singapore, at a cost below S\$100/tCO_2e.

1.2 Proposed solution

The proposed configuration integrates a 210 MW_{th} HTGR with an SMR hydrogen-production system (Figure 2). Nuclear fission heats helium in the primary reactor circuit, and an IHX transfers thermal energy to an independent secondary helium loop supplying the reformer. This arrangement replaces fossil-fired reformer heating and keeps the nuclear and chemical systems physically separate [10]. Reactor capacity beyond the process duty covers the MEA reboiler gap and 19.8 MW_e of site electricity, and the remaining 31.5 MW_{th} stays as spare with no revenue.

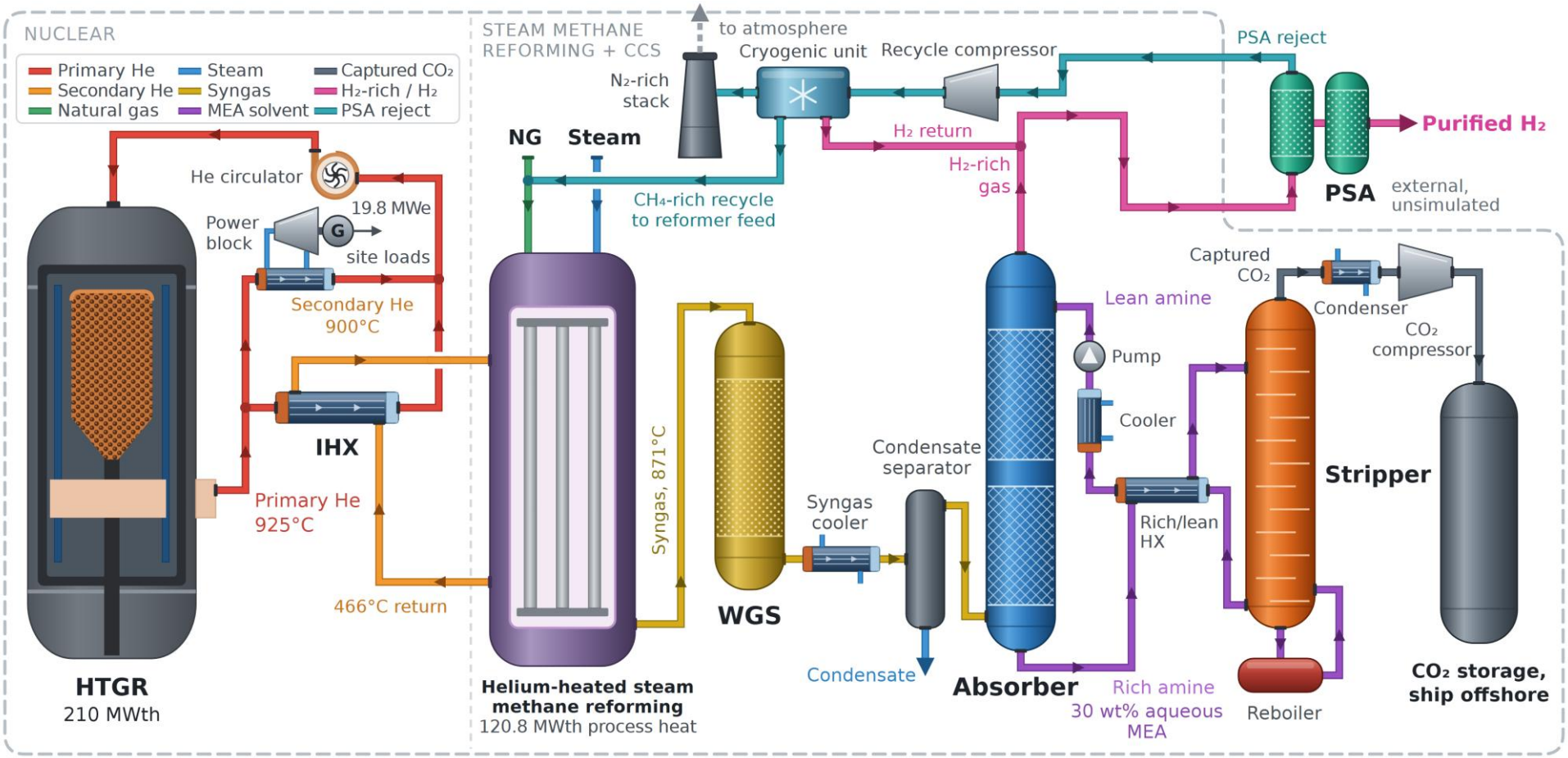


Figure 2. NUS-H_2 arrangement. A 210 MW_{th} HTGR supplies 120.8 MW_{th} of process heat through an IHX and secondary helium circuit. The dashed line is the boundary: CO_2 compression and ship offshore storage sit inside it, the external PSA sits outside, and its reject returns across it.

Natural gas and steam react over a nickel catalyst in the helium-heated reformer to produce hydrogen-rich synthesis gas [4,11]. Water-gas shift (WGS) conversion increases hydrogen yield and converts carbon monoxide to CO_2. After syngas cooling and condensate removal, a 30 wt% aqueous MEA system absorbs CO_2 from the hydrogen-rich gas. A stripper regenerates the rich solvent, releasing concentrated CO_2 for compression and shipment to offshore storage. The Idaho National Laboratory (INL) reference removes CO_2 with Selexol [6,11], so MEA regeneration energy is assessed separately from INL's utility data. Pilot tests of 30 wt% MEA on flue gas report specific reboiler duties of approximately 3.46 to 4.0 GJ tCO_2^{-1} [12,13]. The lower value excludes heat loss at 26 to 27 wt%, the upper value is at 85% capture on 3.5 vol% CO_2, and no measurement on syngas was found.

The PSA stays an unsimulated external block. Its reject returns across the boundary through a recycle compressor and a cryogenic unit, as in INL's Case 6 [6]. Methane-rich gas goes back to the reformer feed, hydrogen returns ahead of the PSA, and a nitrogen-rich cut is burned.

1.3 Design basis

Table 1 sets the headline Case B beside the INL reference Case A. Case B scales the extensive items of INL's 130 MMSCFD Case 6 by 0.6833 to the SOXAL rate, with the reactor resized. Hydrogen reaching the PSA is 10,214 kg h^{-1}. At the published 88% recovery, 8,988 kg h^{-1} leaves as purified product and 1,226 kg h^{-1} returns in the reject. Annual output counts the 85% availability hours only (7,446 h y^{-1}).

Table 1. Headline Case B at the SOXAL rate beside the INL reference Case A, at 85% availability.

Quantity	Case B, headline	Case A, INL Selexol	Case A, with MEA
H_2 product, kg h^{-1}	8,988	13,154	13,154
Annual H_2, t H_2 y^{-1}	66,925	97,946	97,946
Natural gas, MMSCFD	23.2	34.0	34.0
Process heat, MW_{th}	120.80	176.80	176.80
Process electricity, MW_e	12.33	17.30	18.05
Helium circulators, MW_e	7.45	10.44	10.89
Reactor needed, MW_{th}	178.49	242.41	260.74
Rating, MW_{th}	210.00	210.00	210.00
Surplus or shortfall, MW_{th}	31.51	−32.41	−50.74
Captured CO_2, tCO_2 y^{-1}	359,777	542,362	526,544
Net abatement, tCO_2e y^{-1}	578,052	861,812	845,994

2. Technical and Practical Feasibility

2.1 Process heat and integration

The proposed system combines established SMR and CO_2-capture operations with continuous high-temperature heat. Helium heats the reformer directly, so no natural gas is burned for reforming heat, and the chemical plant retains established reforming and amine equipment. Japan's High Temperature

Engineering Test Reactor reached a 950°C outlet [14], yet it is a 30 MW_t test reactor. China's HTR-PM entered commercial operation in December 2023 [15] and runs near 750°C [16]. This work identified no commercial reactor at 900°C, so the design assumes a generic 925°C HTGR and treats Xe-100, also near 750°C [17], as a labelled analogue only. Secondary helium reaches the reformer at 900°C and returns at 466°C [6]. The 871°C reformer outlet is INL's equilibrium assumption. Run through the Level 3 flowsheet, a cooler outlet leaves more methane unconverted: at 825 and 800°C abatement falls to 0.559 $MtCO_2e\ y^{-1}$ and 0.544 $MtCO_2e\ y^{-1}$ (Table 5).

2.2 Thermal and electrical balance

Table 2. Heat and power balance, Case B central. Every row is an engine row, and the heat column sums to the reactor need with no residual. Purge recovery is Case 6 fuel-gas recycle and nitrogen removal equipment, split not published; INL's 5.7 MW_e is at Case A [6]. Reboiler steam passes a back-pressure stage first, so the extraction work and the turbine heat sum to less than the power generated divided by 39.7%.

Heat term	MW_{th}	Electricity term	MW_e
Process heat to the reformer	120.80	Pump, compressors and purge	9.44
CO_2 slip sensible heat	0.88	Cooling towers and MEA fans	0.83
IHX loss	0.97	Reforming loads	0.82
Duct loss	0.31	Purification loads	0.01
Vessel loss	2.66	Steam system loads	0.20
Reactor steam to the reboiler	13.12	Water system loads	1.02
Extraction work, reboiler steam	2.22	Tail-gas power	0.00
Heat to the turbine	44.23	Helium circulators	7.45
Less circulator shaft heat	−6.70		
Reactor needed	**178.49**	**Generated on site**	**19.78**
Rating	210.00	Unmet electricity	0.00
Spare, no revenue	31.51	Unmet heat at rating	0.00

Table 2 closes the balance at the headline basis term by term, with duct loss, vessel loss and circulator shaft heat as separate rows. Starting from INL's rate with no losses, the sizing walk of Table A2 counts the purge load, heat losses, helium circulators and the reboiler gap and reaches 179.0 MW_{th} on the anchored terms. The headline runs the syngas heat recovered above 130°C on its gated model, which passed its benchmark gate, and this lowers the need by 0.55 MW_{th} to 178.5 MW_{th}, leaving 31.5 MW_{th} of the rating spare with no revenue. The spare depends on the heat basis: on Ahn and Lee's 143.4 MW_{th} of process heat [18] it falls to 6.5 MW_{th}. The adverse corner of the sizing range needs 214.1 MW_{th}, 4.1 MW_{th} above the rating; held at the rating, the heat duties are still met and 1.6 MW_e is imported. Case A with MEA and recycle needs 260.7 MW_{th}, so it stays a reference and is never printed as a 210 MW_{th} plant.

2.3 MEA carbon capture

The design uses 30 wt% MEA as the reference solvent, with Selexol and activated MDEA (aMDEA) as sensitivities. The literature is mixed: IEAGHG 2022-07 calls MEA the preferred chemical solvent

[19], while IEAGHG 2017-02 favours aMDEA on shifted syngas [5]. The absorber takes syngas at 40°C from the last knock-out drum and runs at 26.8 bar with rich loading capped at 0.50 mol mol^{-1}, and per-pass capture of 90% is a project assumption. A rate-based absorber solved to that duty needs 15.16 m of packing at Case B. The specific reboiler duty (SRD) of 3.8 GJ tCO_2^{-1}, range 3.4 to 4.2, is assumed from flue-gas pilots, since no syngas measurement exists. The equilibrium-stage stripper overstated the DICE pre-FEED design SRD by 46% and the Kwak pilot SRD by 27% [20,21], so the column gate failed and the assumed SRD stays in the headline (Appendix E). The 51.0 MW reboiler at 121°C draws first on freed Selexol steam (12.7 MW), then diverted export steam (16.8 MW), syngas heat above 130°C (8.4 MW) and reactor steam for the remaining 13.1 MW. The column models rest on the Aronu extended UNIQUAC solvent model [22].

2.4 Reactor safety and emergency planning

Tristructural isotropic (TRISO) fuel particles incorporate multiple protective layers that retain fission products, while helium cooling and the graphite-based core support high-temperature operation [10]. The primary helium boundary, IHX and independent secondary loop provide successive physical barriers between the reactor and hydrogen-production equipment.

Singapore's compact geography makes site-specific emergency planning essential [23]. Advanced-reactor assessments use accident frequencies and projected radiological consequences to determine protective actions [24,25,26]. Figure A1 illustrates this method using the Xe-100 safety-case example [27]. That assessment presents a 400 m emergency-planning reference distance for its specific case [27,28], and HTR-PM studies evaluate offsite arrangements [29,30]. Here 400 m is a conceptual reference only and is no approved Singapore boundary. Emergency arrangements would come from a reactor-specific safety case and regulatory review.

2.5 Regulation and community engagement

Singapore is strengthening its capability to assess advanced nuclear technologies and has made no decision on nuclear energy [31,32]. A Phase 1 International Atomic Energy Agency (IAEA) Integrated Nuclear Infrastructure Review (INIR) is tentatively set for 2027 and covers 19 areas of infrastructure, including emergency preparedness and radioactive waste management [31]. Manpower schemes for public officers begin in 2027 [33], and separately NUS and NTU are working towards a joint Master's in nuclear engineering in 2027 [34]. The Singapore Nuclear Research and Safety Institute contributes reactor modelling and safety analysis alongside public education [32,35,36]. The National Environment Agency's responsibilities and the Energy Market Authority's feasibility study add to the institutional basis, supported by regulatory cooperation abroad [37,38,39]. A future project needs one licensing framework that spans the reactor and its industrial heat interface, with emergency preparedness built in.

Early engagement with Western Island planners, neighbouring Jurong Island operators and nearby communities would support informed assessment [35,36,40]. Clear communication of the plant's industrial purpose and protective design features is central to project development.

2.6 Risks and limitations

The assessment adopts INL process conditions [6,11] and a 210 MW_{th} rating, with balances anchored on INL cases and one term on a gated model (Appendix E). IHX performance and the transfer of INL's recycle to MEA need integrated verification, as does the 39.7% power-conversion assumption. Table 3 summarises the principal constraints.

Table 3. Principal risks and constraints.

Area	Risk or constraint	Mitigation or requirement
Technology	No commercial nuclear process heat for hydrogen; IHX unqualified for 900 to 925°C helium.	Overseas demonstration and component qualification before any Singapore commitment.
Reformer hot end	INL's 29 K helium-to-process approach at the reformer is untested; JAERI's reformer tube rig ran 102 to 125 K [41].	Qualify a helium-heated reformer tube; at an 800°C outlet abatement falls to 0.544 Mt y^{-1}.
Recycle with MEA	At Case A, 179 kmol h^{-1} of CO_2 slip enters the cryogenic unit with MEA, against 17 kmol h^{-1} in INL's Selexol case, unmodelled by INL [6].	Pilot the recycle; without it abatement is 0.467 Mt y^{-1}.
Outage and backup	No furnace, so hydrogen stops in reactor outages.	Offtake for 85% of hours, or a priced backup furnace.
Safety and siting	A dose-based, design-specific EPZ must be justified on a compact site.	Safety case and EPZ analysis before site approval.
Regulation	No licensing pathway exists for a reactor supplying industrial heat.	INIR review, then one coordinated pathway.
Economics	Central full cost is S$165.6/t$CO_2$e to S$277.9/tCO_2e, above the target.	Overseas replication and contracted costs before investment (Section 4.7).
Storage and offtake	Storage depends on Singapore-Indonesia cooperation; offtake is uncontracted.	Contracted CO_2 chain and hydrogen offtake before investment.
Storage accounting	Treatment of offshore storage across the border is unresolved.	Settle under the Singapore-Indonesia arrangement [42].

2.7 Location and roadmap

The proposed Western Island is a siting proposal of this study and has no official approval. The government has announced that smaller islands around Jurong Island will be connected into a new western island, reclaimed over coming decades, and its page names new power generation infrastructure with no mention of nuclear [43,44]. The reference case is a new build at SOXAL scale, so it retires no existing source. Central hydrogen runs only in the 85% availability hours, and backup firing stays a sensitivity. Captured CO_2 is shipped to offshore storage abroad under Singapore-Indonesia cooperation [42]. Pipeline links to the island are unpriced.

Figure 3 sets out an indicative roadmap to potential operation by 2050, and every date is illustrative. National capability and overseas reactor and IHX qualification come before Singapore-specific site assessment, licensing, investment and construction, and each gate opens only once its requirements are met. Land may not exist at the 2039 milestone, since reclamation has no published date. Stakeholders also include the Economic Development Board and CO_2 storage partners. The roughly 11 years between HTR-PM construction start and commercial operation give one historical reference for the schedule [15].

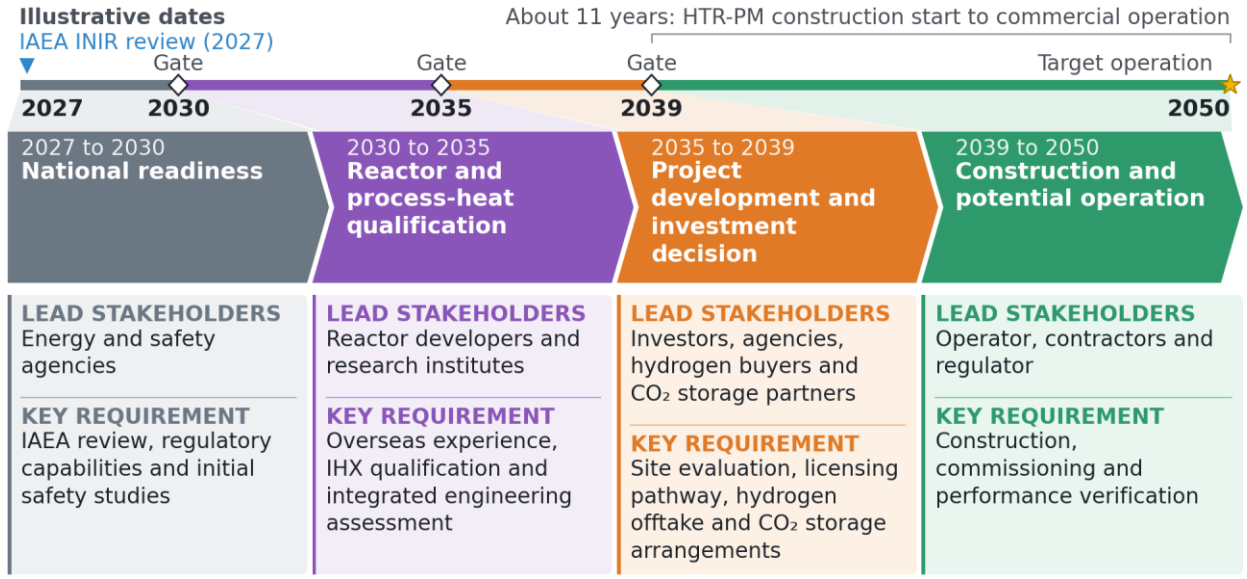


Figure 3. Indicative roadmap with milestones, stakeholders and enabling requirements. All dates are illustrative.

3. Net Emissions Abatement

3.1 Method

Net abatement is the difference between the operational emissions of a same-size unabated SMR and the carbon the proposed plant sends to air:

$$\Delta E_{\text{net}} = E_{\text{baseline}} - E_{\text{residual}} \tag{3.1}$$

$$E_{\text{residual}} = E_{\text{cut}} + E_{\text{grid}} \tag{3.2}$$

The furnace is removed, yet the nitrogen-rich cut of recycled PSA reject is burned, and that carbon E_{cut} is the residual. Absorber slip returns through the reformer and reaches air only through this cut. The grid term is zero as a calculated result at central inputs: unmet electricity is 0 MW_e at the rated reactor. All values are in tonnes per year at 85% availability.

3.2 System boundary and assumptions

The boundary follows Figure 2. It holds a nuclear island (reactor, primary and secondary helium circuits, small power block) and a chemical plant (reformer, WGS, MEA capture, recycle compressor, cryogenic unit, CO_2 compressor and ship offshore storage). The PSA sits outside the line, with its

reject returning across it. Ship and storage sit inside the line for mass balance and cost, while ship fuel stays in the lifecycle sensitivity. The CO_2 compressor load is INL's pipeline basis scaled to Case B, and the liquefaction a ship needs is neither modelled nor priced.

The baseline uses SOXAL's scale of about 100,000 $Nm^3\ h^{-1}$ [8]. At 0.08988 kg Nm^{-3} and 7,446 h y^{-1}, output is 66,925 t $H_2\ y^{-1}$. Baseline intensity is INL's 9.21 kg $CO_2\ kg^{-1}\ H_2$ [11], with 9.35 from Cho et al. [45] and 8.996, the unrounded IEAGHG Base Case value [5], as bounds. Per-pass capture of 90%, a project assumption, gives 90.4% plant capture because slip returns to the reformer, so the two always print together.

3.3 Net emissions abated

Table 4 gives the carbon ledger. Abatement is 578,052 $tCO_2e\ y^{-1}$ (0.578 $MtCO_2e\ y^{-1}$), a 93.8% reduction that takes direct intensity from 9.21 to 0.57 kg $CO_2\ kg^{-1}\ H_2$ and stands at 2.3 times the target. As a size comparison only, it equals 1.2% to 1.3% of Singapore's 2035 target level of 45 to 50 $MtCO_2e$ [46].

Table 4. Carbon ledger, Case B, direct emissions to atmosphere on the INL baseline.

Line	$tCO_2e\ y^{-1}$
Same-size unabated SMR baseline	616,361
Less gas saved by nuclear heat and recycle	216,825
Feed carbon, fresh natural gas	399,536
Captured and stored	359,777
Carbon in product methane, not counted at the plant	1,450
To atmosphere, burned nitrogen-rich cut	38,309
Net abatement, baseline less atmosphere	**578,052**

3.4 Sensitivity and interpretation

Table 5. Net abatement sensitivities, $MtCO_2e\ y^{-1}$. The target is 0.25.

Case	Result	Case	Result
Per-pass capture 80%	0.564	Baseline, Cho 9.35	0.587
Per-pass capture 85%	0.571	Baseline, IEAGHG 8.996	0.564
Per-pass capture 95%	0.584	Ship fuel burden counted	0.569
Availability 65%	0.442	Lifecycle, burdens only	0.566
Availability 75%	0.510	Lifecycle, with saved upstream gas	0.616
Availability 90%	0.612	System view, with lost steam	0.528
Selexol reference	0.589	No recycle, tail gas burned	0.467
aMDEA	0.589	Capture offline	0.218
Reformer outlet 825°C	0.559	Condensate CO_2 vented	0.576
Reformer outlet 800°C	0.544		

Table 5 tests the result. Every case with working capture and recycle stays above the target. Two dependencies matter most. With capture offline, the only abatement left is carbon never generated, 0.218 $MtCO_2e\ y^{-1}$, which is below the target. If recycle fails to transfer from INL's Selexol case to

MEA, gas rises to 30.4 MMSCFD and abatement falls to 0.467 $MtCO_2e\ y^{-1}$. A cooler reformer outlet leaves more methane unconverted, and more carbon then leaves in the burned nitrogen-rich cut.

Backup firing over the 1,314 h y^{-1} of outage would add 11,810 t $H_2\ y^{-1}$. An unabated backup train emits 108,770 $tCO_2e\ y^{-1}$, which matches the unabated SMR in those hours and adds no abatement, while a backup train with capture would add 79,753 $tCO_2e\ y^{-1}$; neither train's capital is priced. Abatement is counted at the Singapore source, storage abroad follows the Singapore-Indonesia memorandum [42], and how offshore storage counts across the border remains unresolved.

4. Economics and Implementation

4.1 Method and boundary

Two methods feed one result, both in real 2025 Singapore dollars at 8%, with S$1.3065/US$ [47] and escalation factors of 1.6195 from 2019 US dollars and 2.322 from 2014 euros [48,49]. The first is an annual incremental ledger against a same-size new-build unabated SMR at 66,925 t $H_2\ y^{-1}$, with levels L0 to L5 adding items step by step:

$$\Delta C = \sum_i C_i, \quad \mathrm{AC} = \frac{\Delta C}{A}, \quad \mathrm{CRF} = \frac{r}{1-(1+r)^{-L}} \tag{4.1}$$

Here C_i is an annual cost line, A is net abatement and CRF annualises capital over life L at rate r, giving 0.0937 at 25 y and 0.0808 at 60 y. Common items cancel, including reformer capital differences by assumption. The target sets a ceiling of S$57.8M/y.

The second is a year-by-year discounted cash flow (DCF) of the unabated SMR and NUS-H_2 at equal output on a common horizon:

$$\mathrm{LCOH}_x = \frac{\sum_t c_{x,t}(1+r)^{-t}}{\sum_t m_t(1+r)^{-t}}, \quad \mathrm{LAC} = \frac{\sum_t \Delta c_t(1+r)^{-t}}{\sum_t A_t(1+r)^{-t}} \tag{4.2}$$

Reactor spend runs uniformly over 5.5 y [50], then operation runs at 85%. Items with a 25 y life are replaced on longer horizons with unused life credited at the end, and decommissioning is paid from cash flow at end of life. Both plants carry the SMR island of the IEAGHG Base Case, S$517.6M of capital and S$17.5M/y of fixed O&M [5]. The two methods share inputs and differ in timing, so their agreement tests the annualisation and does not validate the inputs.

4.2 Reactor capital

INL's recipe removes the 9% conversion share and converts at 40% efficiency [51]:

$$c_{\mathrm{BOAK}} = 6{,}000 \times 0.91 \times 0.40 = 2{,}184\ \mathrm{US\$_{2019}/kW_t}, \quad c_{\mathrm{FOAK}} = 1.6\, c_{\mathrm{BOAK}} \tag{4.3}$$

BOAK is INL's term for a build between a first and Nth of a kind. Central capital uses $n = 1.0$, which charges INL's unit cost at any size: S$743M at BOAK and S$1,188M at FOAK. Two size treatments

print beside it at equal prominence, with BOAK capital of S$1,130M and S$1,532M. Here $n = 0.6$ is a project sensitivity, and its 600 MW_{th} reference size is a project assumption. The $n = 0.31$ value is the ratio of two single-unit FOAK estimates from INL, so it carries that label [51]. Learning at 10% per doubling reaches a tenth unit by two published routes: US$2,462/$kW_t$ from FOAK and US$1,539/$kW_t$ from BOAK [51].

4.3 Capture, storage, integration and operating costs

Capture capital is S$2.04M per t/h from IEAGHG Case 1A [5], with fixed O&M at an assumed 2.7% of capital. Case 1A is an aMDEA case, and Table 9 carries the MEA flue-gas alternative. Capture energy sits in the heat and power balance, so no energy line appears and nothing is counted twice. MEA make-up and ship liquefaction have no cost line. CO_2 transport and storage (T&S) is S$60/t$CO_2$, with 23.2 low [5] and 112 high [52]; no published Singapore-Indonesia tariff was found [53,54]. Gas is S$15/GJ [55], and the plant saves 12.6 MMSCFD against the unabated SMR. Reactor O&M and fuel is US$10/$MWh_{th}$ [51]. IHX and loop, licensing and decommissioning are labelled proxies, and the first two rest on primary sources not verified here. Reactor life changes only the reactor, licensing and decommissioning lines.

4.4 Headline result and cost ladder

Table 6 prints the headline. Every real plant cost is in, the reactor is charged at 210 MW_{th} and $n = 1.0$. The full cost by ledger is S$189.7/t$CO_2$e and S$165.6/tCO_2e at BOAK, and S$277.9/t$CO_2$e and S$241.7/tCO_2e at FOAK. The DCF lands within 0.2% on every cell of the grid, inside the 10% test. None of the 8 central cases reaches S$100/t$CO_2$e. Hydrogen costs S$1.64/kg more than from the unabated SMR at BOAK, 25 y, which is the premium a buyer would have to carry.

Table 6. Headline rebuilt cost by both methods, $n = 1.0$, 210 MW_{th} charged, against the unabated SMR.

Measure	BOAK, 25 y	BOAK, 60 y	FOAK, 25 y	FOAK, 60 y
Full cost, ledger L5, S$/t$CO_2$e	**189.75**	**165.64**	**277.93**	**241.70**
Full cost, DCF LAC, S$/t$CO_2$e	**190.06**	**165.97**	**278.43**	**242.19**
DCF less L5, %	0.16	0.20	0.18	0.20
Over the target, L5, S$/t$CO_2$e	89.75	65.64	177.93	141.70
LCOH, NUS-H_2, S$/kg	5.26	5.05	6.02	5.71
LCOH, unabated SMR, S$/kg	3.62	3.62	3.62	3.62
LCOH premium, S$/kg	1.64	1.43	2.40	2.09
Named step L3, S$/t$CO_2$e	137.70	117.24	209.92	179.54

Table 7 shows what each item adds. The gas saved offsets most of the reactor capital line, and T&S and reactor O&M then take the cost past the target. L3 is a named step, S$137.7/t$CO_2$e and

S$117.2/tCO$_2$e, and it omits interest during construction, the power island, the recycle train and lost steam, all real costs of this plant.

Table 7. Cost ladder, annual lines in S$M/y against the unabated SMR, n = 1.0, 210 MW$_{th}$ charged, 0.578 MtCO$_2$e y^{-1}. IHX, licensing and recycle train are proxies.

Level	Line	BOAK, 25 y	BOAK, 60 y	FOAK, 25 y	FOAK, 60 y
L0	Gas saved	−64.80	−64.80	−64.80	−64.80
L0	Reactor capital	69.58	60.01	111.33	96.02
L0	Capture capital	9.25	9.25	9.25	9.25
L0	Capture fixed O&M	2.67	2.67	2.67	2.67
L0	CO_2 transport and storage	21.59	21.59	21.59	21.59
L1	Reactor O&M and fuel	25.32	25.32	25.32	25.32
L2	IHX and secondary loop	8.36	8.36	8.36	8.36
L3	Extra licensing	6.12	5.28	6.12	5.28
L3	Decommissioning	1.50	0.09	1.50	0.09
L3	**Named step, S$M/y**	**79.60**	**67.77**	**121.34**	**103.78**
	Named step, S$/tCO$_2$e	**137.70**	**117.24**	**209.92**	**179.54**
L4	Interest during construction	13.75	11.86	22.01	18.98
L4	Power island	1.62	1.40	2.59	2.24
L4	Recycle and cryogenic train	5.89	5.89	5.89	5.89
L5	Lost steam export	15.23	15.23	15.23	15.23
L5	Baseline purchased power avoided	−6.41	−6.41	−6.41	−6.41
L5	**Full cost, S$M/y**	**109.69**	**95.75**	**160.66**	**139.71**
	Full cost, S$/tCO$_2$e	**189.75**	**165.64**	**277.93**	**241.70**

4.5 Uncertainty

Table 8 holds the three size treatments at equal weight, and Figure D1 draws them. Table 9 varies one input around BOAK, 25 y, n = 1.0; corner runs carry no probability. The transfer of recycle and the reactor cost treatment move the result most, and availability comes next. A discount rate of 5% or 10% gives S$136.4/tCO$_2$e or S$231.4/tCO$_2$e, a swing close to that of the gas saving, where INL's 35.2% and Ahn and Lee's 18.8% remain unreconciled [6,18]. SRD moves the spare at a fixed 210 MW$_{th}$ and leaves the cost almost unchanged. Charging only the 178.5 MW$_{th}$ the process needs gives S$161.2/tCO$_2$e and S$140.4/tCO$_2$e and removes the design margin.

Table 8. Rebuilt full cost L5 in S$/tCO$_2$e at 25 y / 60 y, 210 MW$_{th}$ charged, for the three size treatments.

Stage	n = 1.0	n = 0.6	n = 0.31
FOAK	277.9 / 241.7	411.0 / 358.2	548.8 / 478.8
Tenth unit, from FOAK	208.5 / 181.8	306.0 / 267.7	407.0 / 356.6
BOAK	189.7 / 165.6	277.7 / 243.2	368.7 / 323.6
Tenth unit, from BOAK	146.3 / 128.2	212.1 / 186.6	280.1 / 247.1

Table 9. One-at-a-time changes to full cost L5, S$/tCO$_2$e, around 189.7 at BOAK, 25 y, n = 1.0. Every cell is an engine sweep row. The gas-saving row holds captured and emitted carbon at central values, and the Case 3 row takes IEAGHG's MEA capture capital.

Change	Result	Change	Result
T&S S$23.2/tCO$_2$	166.85	T&S S$112/tCO$_2$	222.11
Gas S$10/GJ	218.33	Gas S$20/GJ	161.17
Reactor, INL low	141.69	Reactor, INL high	213.78
Availability 65%	252.96	Availability 90%	178.34
Gas saving at 18.8%	242.03	Capture capital, Case 3 MEA	220.52
Discount rate 5%	136.45	Discount rate 10%	231.40
SRD 3.4	189.72	SRD 4.2	189.77
Selexol reference	176.32	aMDEA	187.84
Baseline, Cho	186.72	Baseline, IEAGHG	194.56
Reformer outlet 825°C	204.78	Reformer outlet 800°C	217.00
Reactor sized to need	161.16	No recycle, 0.467 Mt y^{-1}	283.11

4.6 Steam methane reforming plus CCS comparator

One labelled comparator is IEAGHG Case 3, an SMR with MEA on flue gas at 90.0% overall capture, scaled to 100,000 Nm^3 h^{-1} [5]. At 85% availability it abates 0.536 MtCO$_2$e y^{-1} against its own 8.996 baseline. Its published cost converts to S$162.1/tCO$_2$e with T&S at EUR 10/tCO$_2$, or S$203.0/tCO$_2$e at this report's S$60/tCO$_2$. At those T&S values NUS-H$_2$ at BOAK, 25 y, gives S$166.8/tCO$_2$e and S$189.7/tCO$_2$e, while FOAK gives S$277.9/tCO$_2$e at central T&S. The row is indicative, since gas price and scale differ (Table D5).

4.7 Closing the gap to the target

At BOAK, 25 y, the full cost is S$109.7M/y, which is S$51.9M/y over the ceiling. Table 10 tests the levers one at a time. No single lever closes this gap within its tested range: gas would need S$30.7/GJ, reactor capital would need US$824/kW$_t$ against INL's low of US$1,456/kW$_t$, and T&S would have to go negative.

Table 10. Value each lever alone would need to reach S$100/tCO$_2$e at BOAK, n = 1.0.

Lever	L5, 25 y	L5, 60 y	L3, 25 y	Reference
Gas, S$/GJ	30.7	26.5	20.0	Central 15, high 20
Reactor, US$/kW$_t$	824	1,031	1,500	Central 2,184, low 1,456
T&S, S$/tCO$_2$	−84.2	−45.5	−0.6	Central 60, low 23.2

The named scenario MATURE_60Y stacks the changes that a mature reactor fleet and dearer gas would bring (Figure 4). A tenth-of-a-kind reactor on INL's BOAK learning route brings the base S$189.7/tCO$_2$e down to S$146.3/tCO$_2$e at 25 y, and a 60 y life takes it to S$128.2/tCO$_2$e. Gas at S$20.7/GJ against the central S$15/GJ then gives S$95.6/tCO$_2$e on the ledger and S$95.9/tCO$_2$e by DCF, below the target. The printed price is the larger of two solves for S$96/tCO$_2e, S20.63/GJ on the ledger and S$20.68/GJ by DCF, rounded up to 0.1; at S$20/GJ, the top of the tested range, the

scenario sits at S$99.6/tCO$_2$e. Dearer gas lowers the abatement cost because NUS-H$_2$ burns 23.2 MMSCFD against 35.9 MMSCFD for the unabated plant, so each rise in price enlarges the credit for the gas saved. No central case meets these conditions, and the base result stays the reference.

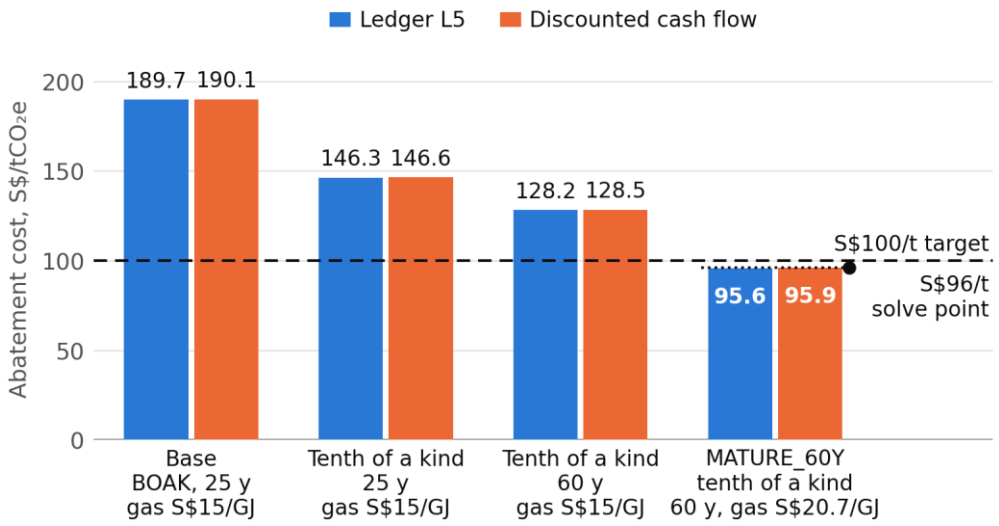


Figure 4. Step-down from the base full cost to the MATURE_60Y scenario. Every bar is an engine row, ledger L5 with the DCF beside it. The dotted line marks the S$96/tCO$_2$e point that sets the printed gas price of S$20.7/GJ, against the central S$15/GJ.

For deployment, the result favours waiting for a design replicated overseas before any Singapore commitment, and replacing proxies with reactor quotations and a contracted T&S tariff. Figure 3 places these steps in time. Carbon tax stays out of the headline because it is a transfer. For an operator, the published rate for 2026 to 2027 and the band stated for 2030 [56] both sit well below the base full cost of S$189.7/tCO$_2$e, so the tax alone would not close the gap. The Future Energy Fund gives a policy context with no allocation assumed [57].

5. Discussion and Conclusion

5.1 Discussion

Environmental performance. The plant abates 0.578 MtCO$_2$e y^{-1} at 90% per-pass and 90.4% plant capture, and the result holds across the baseline bounds. It rests on working capture and on recycle transferring to MEA.

Economic performance. The full cost of S$165.6/tCO$_2$e to S$277.9/tCO$_2$e at $n = 1.0$ misses the target in every central case, by both methods. The MATURE_60Y scenario reaches S$95.6/tCO$_2$e only with a tenth-of-a-kind reactor on INL's BOAK learning route over 60 y and gas at S$20.7/GJ, above the S$10 to S$20/GJ range tested; at S$20/GJ it gives S$99.6/tCO$_2$e.

Comparison with conventional SMR plus CCS. IEAGHG Case 3 at S$203.0/tCO$_2$e on matched T&S sits above the NUS-H$_2$ BOAK case and below its FOAK case, so the proposal competes only

once a reactor is past first of a kind. The comparison is indicative. NUS-H_2 remains distinct by removing the furnace, with residual emissions of 0.57 kg CO_2 kg^{-1} H_2.

Implementation in Singapore. Reactor qualification, the nuclear-chemical heat interface, site-specific emergency planning and contracted CO_2 transport and storage form the main requirements (Figure 3). A competing pathway exists: Air Liquide and Aster have signed a memorandum on autothermal reforming with integrated capture in Singapore [58].

Principal uncertainties. Reactor size treatment spans S$189.7/t$CO_2$e to S$368.7/tCO_2e at BOAK, 25 y. The gas saving, the SRD on syngas, IHX performance and the spare on Ahn and Lee's heat basis all remain unresolved. The column models and the Level 3 flowsheet failed their gates, so most terms stay anchored (Appendix E).

5.2 Conclusion

NUS-H_2 pairs a 210 MW_{th} HTGR with an SMR and 30 wt% MEA at SOXAL scale. The abatement target is met at 0.578 $MtCO_2e$ y^{-1}. The cost target of S$100/t$CO_2$e is missed in every central case, with a base full cost of S$189.7/t$CO_2$e at BOAK, 25 y. It is met only under the stated MATURE_60Y conditions, a tenth-of-a-kind reactor on INL's BOAK learning route over 60 y with gas at S$20.7/GJ, a price above the S$10 to S$20/GJ range tested; at S$20/GJ the scenario gives S$99.6/t$CO_2$e. Singapore's next steps are reactor and IHX qualification overseas, site-specific safety and licensing, hydrogen offtake and a contracted CO_2 storage pathway.

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Appendix A. Reactor design basis

This appendix records the source-traceable design basis and the reactor sizing used in Chapters 1 and 2. Reference values are from INL TEV-961, Table 2, 925°C case [6], with the process configuration of TEV-953 [11]. Sizing rests on balances anchored on these cases and does not demonstrate self-sufficiency of a built plant.

Table A1. Design basis. Case A here is INL's Case 6 with 30 wt% MEA in place of its published Selexol. Published values are from INL TEV-961 [6] and TEV-953 [11]; assumed and calculated values are labelled.

Parameter	Case B	Case A, with MEA	Label
Reactor outlet, °C	925	925	Published
Secondary helium supply, °C	900	900	Published
Secondary helium return, °C	466	466	Published
Secondary helium flow, kg s^{-1}	54.0	79.1	Calculated
Primary helium flow at need, kg s^{-1}	79.2	115.6	Calculated
Reformer outlet, °C	871	871	Published
Process heat, MW_{th}	120.80	176.80	Scaled; published
Electricity demand, MW_e	19.78	28.94	Calculated, with MEA
Power conversion efficiency, %	39.7	39.7	Published, interpolated
PSA recovery, %	88	88	Published, external
Availability, %	85	85	Assumed
Proposed CO_2 capture	30 wt% MEA	30 wt% MEA	Assumed
SRD, GJ tCO_2^{-1}	3.8	3.8	Assumed
Rich and lean loading, mol mol^{-1}	0.50, 0.25	0.50, 0.25	Assumed
Reactor rating, MW_{th}	210.00	210.00	Design decision

Table A2. Sizing walk in MW_{th}. Steps not exported by the results are marked n/e.

Step	Case B	Case A, MEA
INL's rate: process heat plus power at 39.7%, no purge load	206.02	206.02
Same sum at Case B	140.77	n/a
Restore purge load	150.58	220.38
Add vessel, IHX and duct losses	154.27	n/e
Add helium circulators	165.95	242.41
Swap Selexol for MEA, zero reboiler heat	166.25	n/e
Add reboiler gap at 3.8 GJ tCO_2^{-1}, anchored terms	179.04	n/e
Headline, syngas heat on its gated model	**178.49**	**260.74**
Rated	210.00	210.00
Margin	31.51	−50.74

Note 1. INL quotes an equivalent reactor size of 178.8 MW_{th} for Case 6 because it imports 16.5 MW_e [6], so that figure carries almost no site electricity.

Note 2. Losses and the reboiler are counted in the walk, and Case 5 duties stand in for Case 6, whose stream sheets are not held. The walk runs on the anchored terms; its last row is the headline need, with the syngas heat above 130°C on its gated model.

Note 3. The 17.3 MW_e demand belongs to the INL Selexol and PSA flowsheet, including CO_2 compression and purge recovery [6]. The purge row is Case 6 fuel-gas recycle and nitrogen removal equipment, split not published.

Note 4. INL's rate of 29,000 lb/h over 7,446 h y^{-1} gives 97,946 t y^{-1} of product including 0.79 mass% methane. At Case B, 100,000 Nm^3 h^{-1} gives 66,925 t y^{-1}.

The emergency planning zone of Chapter 2 rests on the frequency-consequence method drawn in Figure A1.

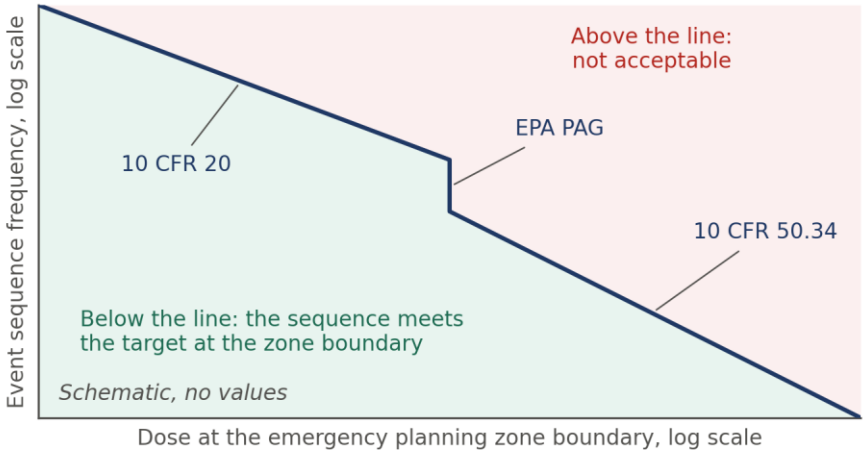


Figure A1. The frequency-consequence method behind an EPZ, drawn as a schematic after the Xe-100 example in the construction permit application [27]. Each accident sequence is placed by its frequency and its 30-day dose at the zone boundary, 400 m in that example, and must fall below the target line. The axes carry no values; the dose criteria and the source figure are in [27].

Appendix B. Mass and energy balance

Stream values come from the integrated balance files, and the PSA streams are external boundary values. The cryogenic unit returns 560 kmol h^{-1} of hydrogen ahead of the PSA, a stream with no carbon. The carbon, hydrogen and mass residuals all print as zero, and the shift releases 8.1 MW.

Table B1. Carbon streams, Case B, from the integrated mass balance, kmol C per hour. A stream crosses the dashed line of Figure 2 when exactly one end sits inside it; the gas supply, the PSA, the product and offsite users sit outside.

Stream	From	To	Crosses	Carbon
Purified H_2 after external PSA	PSA	Product	No	4.4
Compressed CO_2 to storage	CO_2 compressor	Storage	No	1,097.9
Burned nitrogen-rich cut	Cryogenic unit	Stack	No	116.9
Fresh natural gas	Gas supply	Reformer	Yes	1,219.2
Cryogenic H_2 return ahead of the PSA	Cryogenic unit	PSA	Yes	0.0
H_2-rich gas into the PSA	MEA capture	PSA	Yes	476.2
PSA reject returning across the line	PSA	Cryogenic unit	Yes	471.7
Recycled methane-rich gas	Cryogenic unit	Reformer	No	354.8
Tail gas leaving the boundary	PSA	Offsite	No	0.0
Carbon residual				0.0

Table B2. Reboiler heat stack, Case B central, MW, in the order of use.

Source, in order of use	Duty
Freed Selexol steam	12.73
Export steam diverted	16.76
Syngas heat above 130°C	8.39
Reactor steam, the reboiler gap	13.12
Reboiler duty at 3.8 GJ tCO_2^{-1}	**51.00**

Helium circulation at the need is 79.2 kg s^{-1} primary and 54.0 kg s^{-1} secondary, and the circulators draw 7.45 MW_e. The sizing range of 155.8 MW_{th} to 214.1 MW_{th} spans the best and adverse corners of SRD, losses and circulator terms. SRD alone moves the need to 174.1 MW_{th} and 182.9 MW_{th}.

Appendix C. Emissions

The direct method is the headline. Baseline 616,361 $tCO_2e\ y^{-1}$ less residual 38,309 $tCO_2e\ y^{-1}$ gives net abatement of 578,052 $tCO_2e\ y^{-1}$. Lifecycle factors for nuclear power and upstream gas come from the cited sources and were not re-verified against their primary data [59,60]. Ship fuel uses an assumed 0.025 $t\ t^{-1}$ of CO_2 stored.

Table C1. Emissions ledger, Case B on the INL baseline. Lifecycle rows are sensitivities and stay out of the headline.

View	Line	$tCO_2e\ y^{-1}$	$kg\ CO_2\ kg^{-1}\ H_2$
Direct	Unabated SMR baseline	616,361	9.21
Direct	Captured and stored	359,777	5.38
Direct	Residual to atmosphere	38,309	0.57
Direct	Net abatement	578,052	8.64
Lifecycle, burdens	Nuclear lifecycle burden	−2,902	−0.04
Lifecycle, burdens	T&S burden	−8,994	−0.13
Lifecycle, burdens	Net abatement	566,155	8.46
Lifecycle, upstream	Saved upstream gas	49,677	0.74
Lifecycle, upstream	Net abatement	615,832	9.20
System	Net abatement with lost steam	527,632	7.88

Appendix D. Economics

Table entries come from the ledger and DCF results. INL's overnight cost already holds owner costs, licensing and decommissioning shares [51], so the extra licensing and decommissioning lines are labelled proxies that may overlap them.

Table D1. Cost inputs. Label P is published, A assumed, C calculated, U a project decision.

Input	Low	Central	High	Label, source
Reactor overnight cost, US$/kW$_t$	1,456	2,184	2,548	P, C [51]
FOAK premium, multiple of BOAK	1.3	1.6	2.1	P [51]
Reactor O&M and fuel, US$/MWh$_{th}$	6	10	14	P, C [51]
Gas, S$/GJ, higher heating value	10	15	20	P, A [55]
T&S, S$/tCO$_2$	23.2	60	112	U [5,52]
Capture capital, S$M per t/h	n/a	2.04	5.09	C [5]
Capture fixed O&M, % of capital	n/a	2.7	6.0	A, P [18]
IHX and loop line, S$M/y	0.88	8.36	17.24	Proxy, primary not verified
Licensing line, 25 y, S$M/y	0.00	6.12	16.83	Proxy, primary not verified
Decommissioning, US$/kW$_e$	750	1,000	1,250	P [50]
Recycle train line, S$M/y	0.00	5.89	25.35	Proxy from [5]
Discount rate, %	n/a	8	n/a	A [5]
Availability, %	65	85	90	A

Table D2. Reactor capital at 210 MW$_{th}$ and its annual lines.

Stage and size treatment	Capital, S$M	25 y, S$M/y	60 y, S$M/y
BOAK, n = 1.0	742.8	69.58	60.01
BOAK, n = 0.6	1,130.4	105.89	91.33
BOAK, n = 0.31	1,531.5	143.47	123.75
FOAK, n = 1.0	1,188.4	111.33	96.02
FOAK, n = 0.6	1,808.6	169.43	146.13
FOAK, n = 0.31	2,450.5	229.56	197.99
Tenth, from BOAK, n = 1.0	523.4	49.03	42.29
Tenth, from FOAK, n = 1.0	837.5	78.45	67.67

Table D3. Stage, life and size matrix at 25 y / 60 y. Costs are S$/tCO$_2$e, LCOH is S$/kg. Cells marked n/e are not exported.

Case	L3	L5	DCF LAC	LCOH
BOAK, n = 1.0	137.7 / 117.2	189.7 / 165.6	190.1 / 166.0	5.26 / 5.05
BOAK, n = 0.6	213.2 / 184.1	277.7 / 243.2	278.2 / 243.7	6.02 / 5.72
BOAK, n = 0.31	291.4 / 253.4	368.7 / 323.6	369.4 / 324.2	6.81 / 6.42
FOAK, n = 1.0	209.9 / 179.5	277.9 / 241.7	278.4 / 242.2	6.02 / 5.71
FOAK, n = 0.6	323.1 / 278.9	411.0 / 358.2	411.8 / 358.9	7.17 / 6.72
FOAK, n = 0.31	440.3 / 381.8	548.8 / 478.8	549.8 / 479.8	8.37 / 7.76
Sized to need, BOAK, n = 1.0	112.7 / 95.1	161.2 / 140.4	161.4 / 140.7	5.01 / 4.83
Case A at its own need, BOAK	109.1 / 91.9	157.5 / 137.2	n/e	n/e

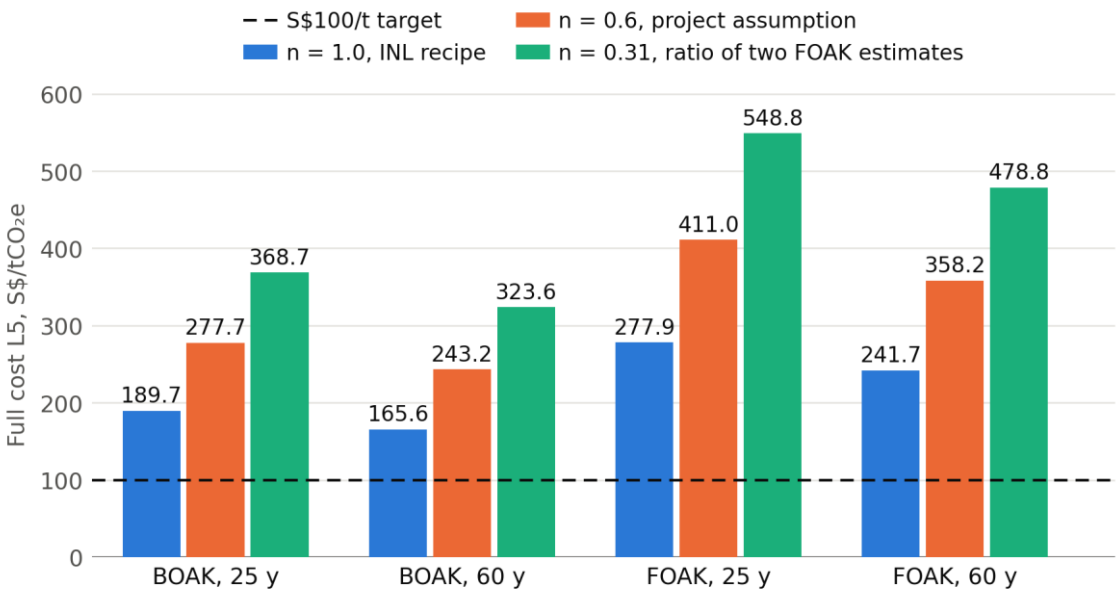


Figure D1. Rebuilt full cost L5 for the three size treatments at equal weight, against the S$100/t$CO_2$e target.

Table D4. Currency and escalation.

Factor	Value
Singapore dollars per US dollar, 2025 average [47]	1.3065
2025 Singapore dollars per 2019 US dollar [48]	1.6195
2025 Singapore dollars per 2014 euro [49]	2.322

Table D5. SMR plus CCS comparator, indicative only, since gas price and scale differ.

Item	IEAGHG Case 3	NUS-H_2, BOAK, 25 y
Capture route	MEA on flue gas	MEA on syngas
Overall capture, %	90.0	90.4
Abatement, $MtCO_2e\ y^{-1}$	0.536	0.578
Baseline, kg CO_2 kg^{-1} H_2	8.996	9.21
Cost at low T&S, S$/t$CO_2$e	162.1	166.8
Cost at S$60/t$CO_2$, S$/tCO_2e	203.0	189.7

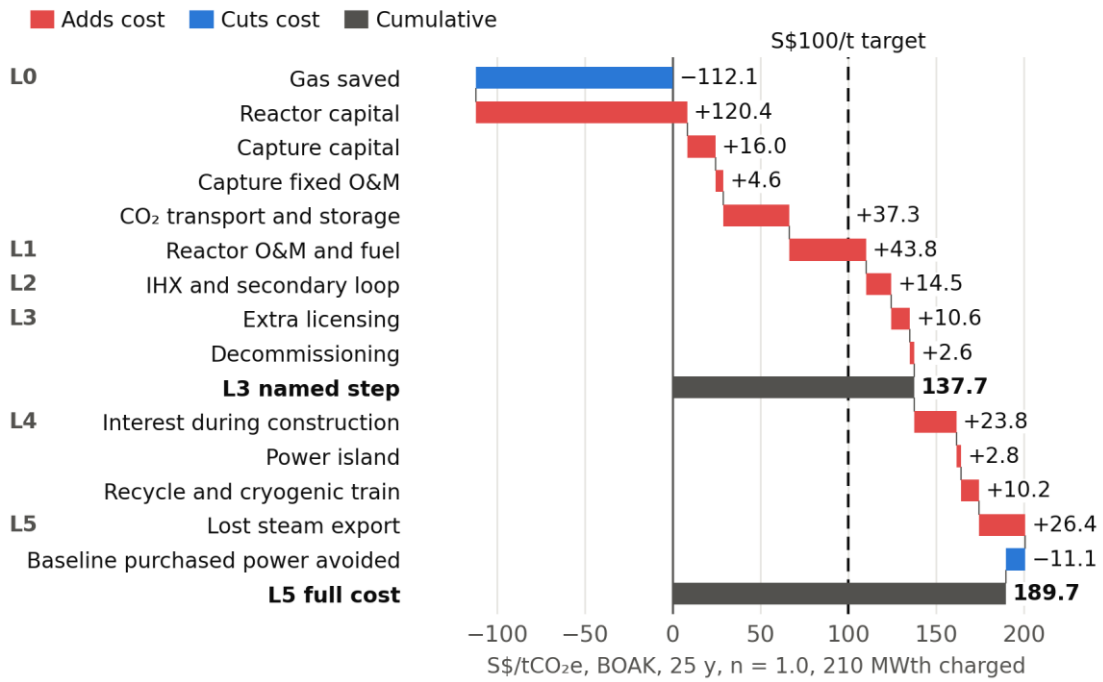


Figure D2. Cost ladder at BOAK, 25 y, in S$/tCO$_2$e, with L3 as a named step.

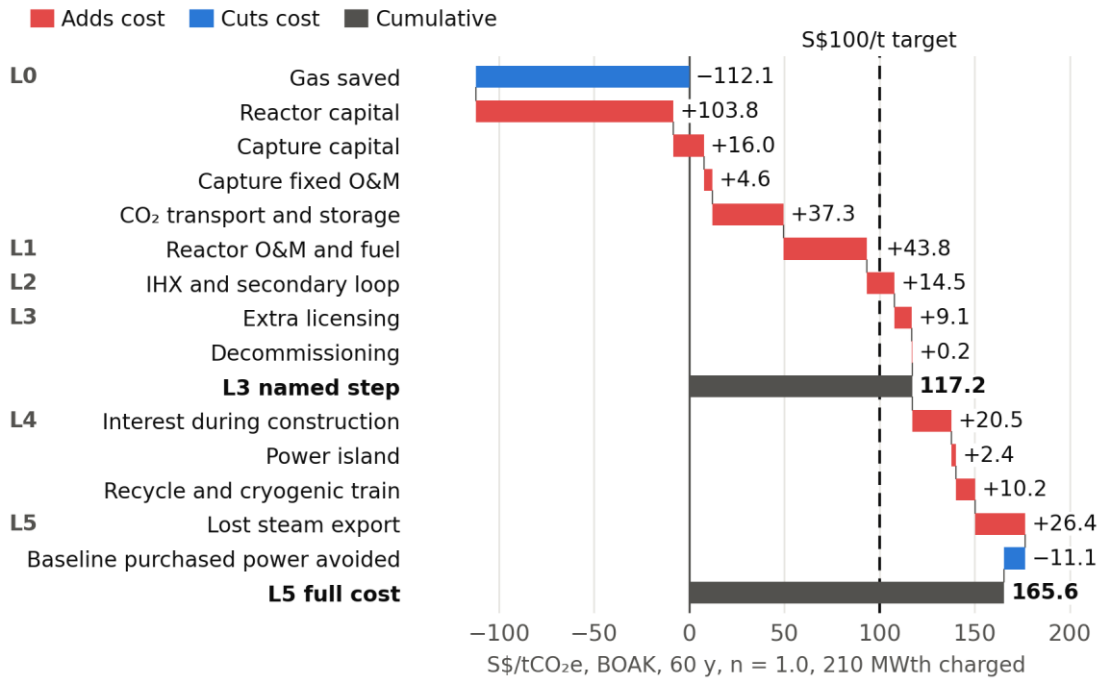


Figure D3. Cost ladder at BOAK, 60 y, in S$/tCO$_2$e, with L3 as a named step.

Appendix E. Model status

This appendix prints only what ran. Each result row carries a status. On the headline preset, term T05, the syngas heat recovered above 130°C, runs on its gated model, and every other term keeps its anchored value: an algebraic balance anchored on INL cases and reproduced in the Rust engine. The rows of the six results tables hold 1,571 anchored rows, 1,800 rows that move with term T05 and 67 rows of the preset that also takes term T01 from its model. Unit models give 2 gas-unit rows and 4 reformer-tube rows, and the column models give 8 rows whose terms keep their anchored values because the column gate failed. The rest are 350 Level 3 flowsheet rows, 99 sensitivity rows that rest on a flowsheet result, 120 heat-recovery rows, 231 input rows and 9 rows not run. PSA streams are external boundary values, and the 39.7% conversion, availability and capture fraction are fixed assumptions.

The validation file records 650 rows: 325 pass, 32 fail, 251 are reported with no limit and 42 did not run. 360 rows use independent data, 269 share a dataset with the model and 21 are calibration, so a pass on a shared dataset verifies code and validates no physics. Table E1 groups the rows by level.

Table E1. Validation rows by level, generated from the validation results. Reported rows print a value with no limit.

Level	Rows	Pass	Fail	Reported	Not run	Independent
1. Properties, equilibria and solvent model	207	189	6	4	8	78
2. Unit benchmarks against published data	345	55	14	247	29	280
3. Level 3 flowsheet	28	20	5	0	3	2
4. Screening reproduction, sizing walk and term gates	47	38	7	0	2	0
5. Cost grid, DCF and headline checks	21	21	0	0	0	0
6. Export and parameter checks	2	2	0	0	0	0
All rows	**650**	**325**	**32**	**251**	**42**	**360**

Table E2 lists the thirteen model terms. A term leaves its anchored value only when its gate passes. The column gate failed on six benchmark rows, among them the high-pressure absorber pilot at 72.6% removal against 85% and the two SRD rows of Section 2.3, so the capture terms stay anchored. The Level 3 flowsheet failed its carbon gate: reformer tube carbon is 4.9% below the anchor and tail-gas carbon 19.7% below, so the recycle, compression and gas terms stay anchored. Two gates could not run: the upstream preheat behind the process heat term rests on Case 6 stream states the library lacks, and no measured helium-side pressure drop exists for the secondary circulator.

Table E2. Model terms on the headline preset. A term uses its model only when its gate passes; the carbon split passed and is used on the modelled preset alone.

Term	Quantity	Gate	Result	Headline value	Source
T01	Carbon split after shift	Gas-train benchmarks	Passed	0.775	Anchored
T02	Per-pass capture	Column benchmarks	Failed	0.9	Anchored
T03	Process heat	Gas-train benchmarks	Not run	120.80 MW	Anchored
T04	Recycle slip heat	Flowsheet carbon	Failed	0.88 MW	Anchored
T05	Syngas heat above 130°C	Gas-train benchmarks	Passed	8.39 MW	Model
T06	Specific reboiler duty	Column benchmarks	Failed	3.8 GJ tCO_2^{-1}	Anchored
T07	Lean pump	Column benchmarks	Failed	0.87 MW_e	Anchored
T08	MEA cooling load	Column benchmarks	Failed	0.77 MW_e	Anchored
T09	CO_2 compression	Flowsheet carbon	Failed	4.25 MW_e	Anchored
T10	Secondary circulator pressure drop	Helium pressure drop	Not run	200 kPa	Anchored
T11	Primary circulator pressure drop	None	No gate	150 kPa	Anchored
T12	Losses	None	No gate	3.95 MW	Anchored
T13	Natural gas	Flowsheet carbon	Failed	23.2 MMSCFD	Anchored

Two open heat terms move the reactor need. The higher secondary helium pressure drop in the sweep raises it to 184.7 MW_{th}, and reading the 39.7% conversion as already net of primary circulation lowers it to 172.0 MW_{th}. Circulator power has no measured basis, and 85% availability has no measured HTGR record behind it.